1. A college freshman must take a science course, a humanities course, and a mathematics course. If she may select any of 4 science courses, any of 3 humanities courses, and any of 2 mathematics courses, how many ways can she arrange her program?

   Science courses = 4
   Humanities courses = 3
   Mathematics courses = 2
   Total ways = Science x Humanities x Mathematics
   = 4 x 3 x 2 = 24 ways
   There are 24 ways that she can arrange her program/courses.

2. A clothing store wants to stock sweatshirts that come in four sizes (small, medium, large, x-large) and in three colors (yellow, blue, and white). How many different types of sweatshirts will the store have to stock?

   Number of Sizes = 4
   Number of Colors = 3
   Total types of sweatshirts to be stocked = 4 x 3
   = 12
   There are 12 different types of sweatshirts that the store will have to stock

3. For her birthday, Karla received a new wardrobe consisting of 5 shirts, 3 pairs of pants, 2 skirts, and 3 pairs of shoes for her birthday. How many new outfits can she make?

   Shirts = 5
   Pairs of pants = 3
   Skirts = 2
   Pairs of shoes = 3

Total outfit she can make = (Shirts x Pants x Pair of Shoes) + (Shirts x Skirts x Pairs of Shoes)

= 5 x 3 x 3
= 45
= 5 x 2 x 3
= 30
= 45 + 30
= 75

Karla can make 75 total outfits combining all of the prowear that was inside the wardrobe